



Executive Summary

Atmospheric Boundary Layer Signal Isolation & Topological Stability

The **SBLToolkit.jl** suite processes complex observational records from nocturnal field tower networks. By leveraging Complex Empirical Orthogonal Functions (CEOF), non-uniform Taylor-Vandermonde derivative operators, and Geometric Surface Pattern Theory (GSPT), the framework cleanly differentiates standing Low-Level Jets (LLJs) from propagating Internal Gravity Waves (IGWs) while mapping saddle-node turbulence collapse manifolds.

CEOF
ACCURACY

99.93%

Wave Wavenumber
Extraction

JET VARIANCE
SHARE

91.37%

Standing Mode 1
Isolation

WAVE VARIANCE
SHARE

8.62%

Propagating Mode 2
Isolation

FOLD SHEAR
 S_{FOLD}

0.6651

Analytical
Bifurcation (s^{-1})

FOLD TKE E_{FOLD}

0.1695

Critical Energy ($m^2 s^{-2}$)

1. Field Campaign Comparative Analysis

Evaluating tower spatial resolution, vertical level density (N_z), and GSPT inversion feasibility.

GSPT transition inversions depend on high vertical tower resolution to construct non-uniform spatial derivative matrices (D1, D2). While multi-level masts like CASES-99 (55m, $N_z=7$) and GABLS3 (200m, $N_z=20$) yield full topological stability mappings, sparse arrays like SHEBA (2 levels) trigger a *Bypassed Guard* due to stencil under-determination.

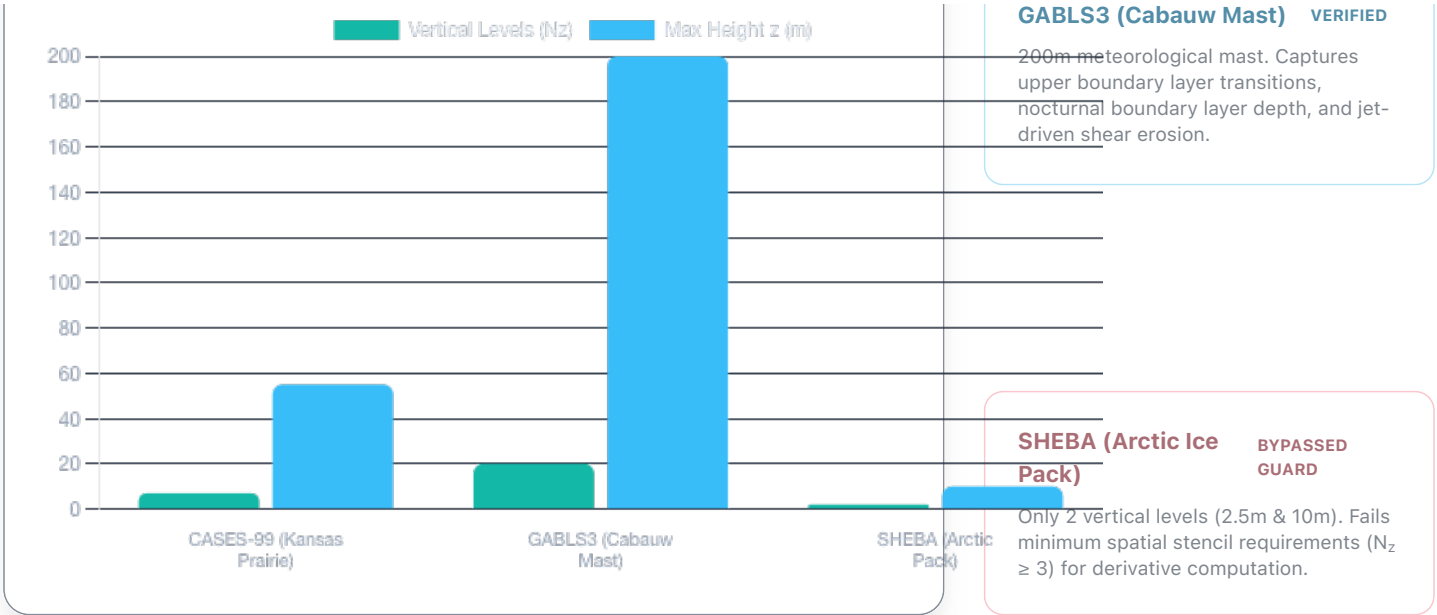


Tower Height & Vertical Grid Density Comparison

CASES-99 (Kansas Prairie)

VERIFIED

55m tower with 7 non-uniform heights (1.5m to 55m). Ideal for capturing nocturnal Low-Level Jet nose formation near 30m.



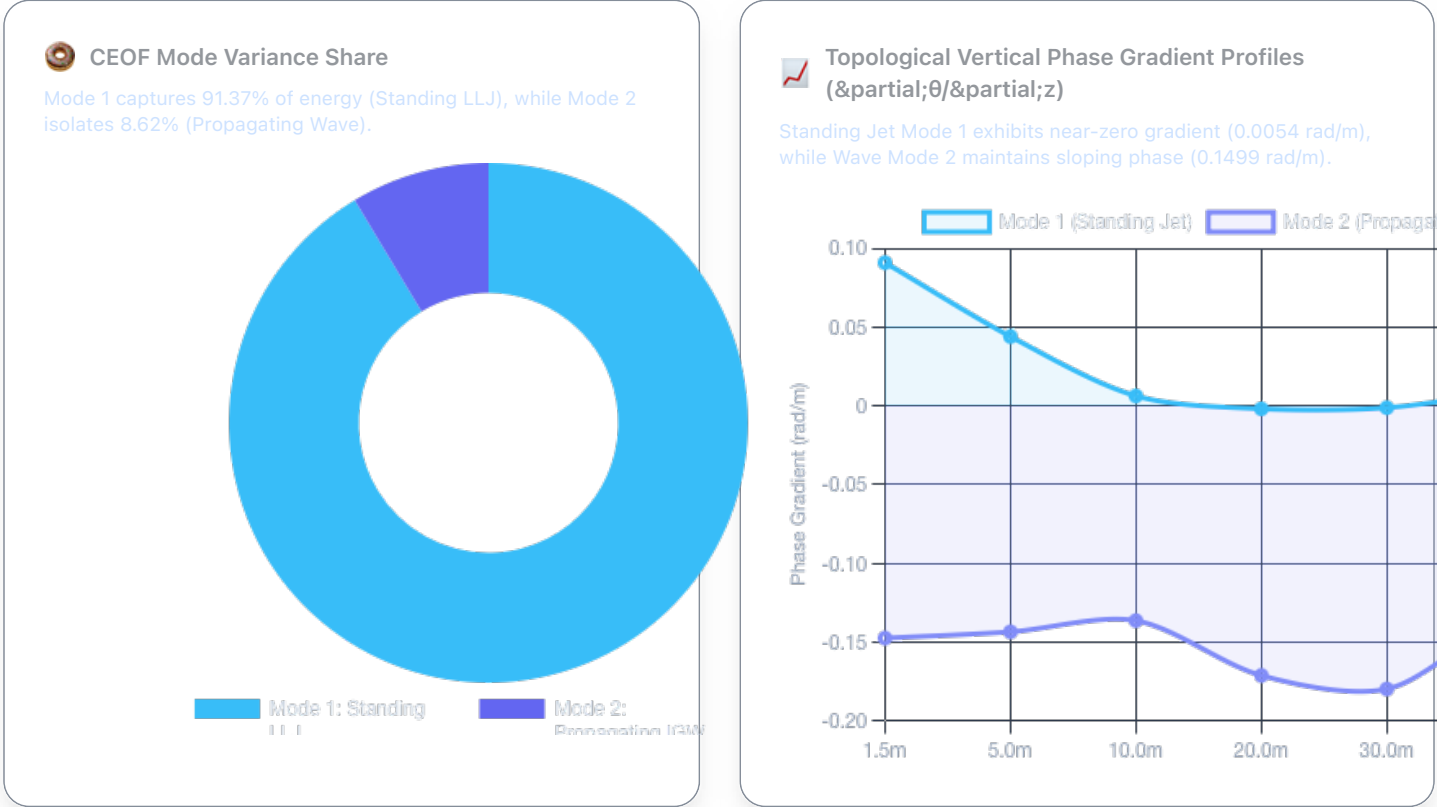


SBL Toolkit & GSPT Framework

Stable Boundary Layer Signal Separation & Topological Bifurcation

[Overview](#) [Field Campaigns](#) [CEOF Wave-Jet](#) [Saddle-Node Dynamics](#)
[Regularization](#) [QC Pipeline](#)

Using a self-contained matrix Fourier Hilbert transform, real velocity fields are converted to complex analytic signals $z(t) = x(t) + iH[x(t)]$. Complex SVD extracts spatial mode amplitudes and vertical phase profiles. Low-amplitude phase noise near the surface is neutralized using an **amplitude-weighted vertical phase gradient metric**.



3. Interactive GSPT Saddle-Node Simulator & Fast-Slow Relaxation Engine

GSPT represents nocturnal boundary layer transitions as a fast-slow dynamical system in 2D phase-space (Shear S vs Turbulent Kinetic Energy e). Solved via depressed-cubic roots analysis, when shear drops below the fold threshold $S_{\text{fold}} = 0.66511 \text{ s}^{-1}$, the stable turbulent equilibrium annihilates, precipitating a rapid collapse to the **Laminar Attractor** ($e \approx 0.0005 \text{ m}^2 \text{ s}^{-2}$). As jet shear slowly rebuilds, the system undergoes periodic relaxation bursts.

Real-Time Numerical Parameter Controls

Adjust geostrophic forcing, relaxation times, and mixing length to recompute trajectory paths and critical fold bounds.

S_{fold} : 0.66511
 s^{-1}

e_{fold} : 0.16954
 $\text{m}^2 \text{s}^{-2}$

Geostrophic Forcing (G_0): 0.44

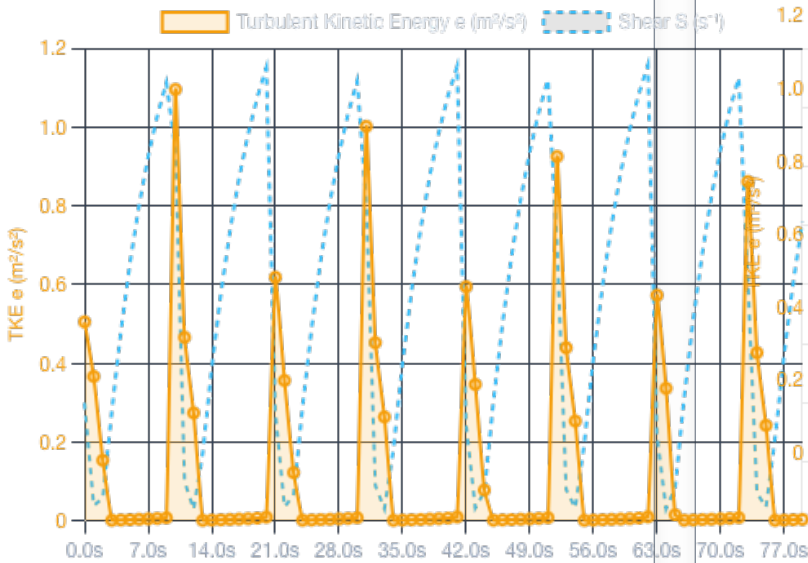
Relaxation Parameter (ϵ): 0.03

Mixing Length (l_0): 0.55



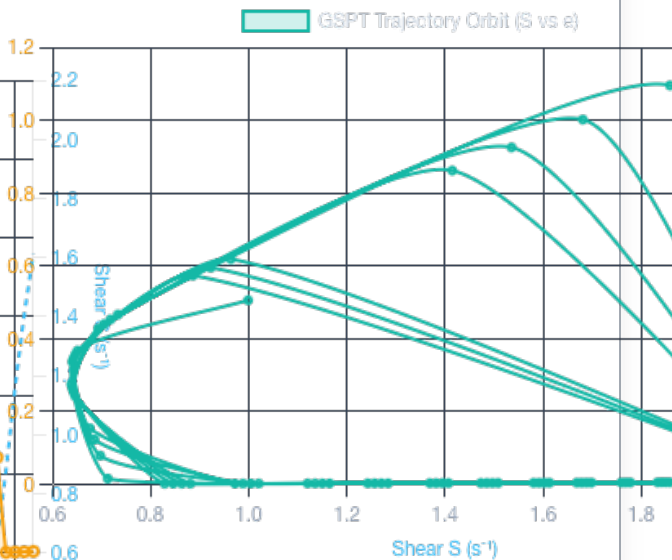
Time Series Trajectory: Shear (S) & TKE (e) Relaxation Cycle

Forward time integration showing sudden TKE collapses and slow wind shear rebuild phases.



2D Phase Portrait: Shear (S) vs. TKE (e) Orbit

Phase-space trajectory cycling around the critical fold line S_{fold} into the laminar regime.



DEPRESSED-CUBIC FOLD ANALYSIS

- Critical Fold Shear: 0.66511 s^{-1}
- Critical Fold TKE: $0.16954 \text{ m}^2 \text{s}^{-2}$
- Physical Mechanism: Canonical 3-real-roots branch selection for positive-shear manifold.

HYSTERETIC STATE PARTITION

- **Stable Turbulent State:** $\text{TKE } e > 0.45 \text{ m}^2 \text{s}^{-2}$ (Active shear mixing)
- **Laminar Attractor:** $\text{TKE } e < 0.01 \text{ m}^2 \text{s}^{-2}$ (Turbulence collapse state)

JET SHEAR REVIVAL CYCLE

During laminar collapse, friction drops, allowing the Low-Level Jet to accelerate. Shear S accumulates until exceeding the upper ignition threshold, triggering intermittent turbulent bursts.

4. Numerical Stability & Singular Value Mitigation

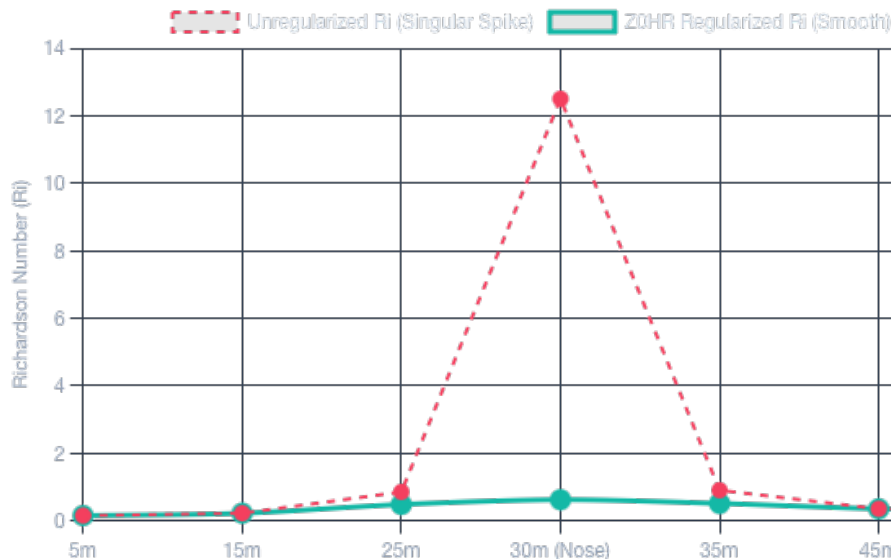
Preventing runaway cooling bifurcations and division-by-zero quotient singularities near Jet cores.

Under strong nocturnal cooling, the gradient Richardson number $Ri = (g/\theta_v \cdot \partial \theta_v / \partial z) / [(\partial u / \partial z)^2 + (\partial v / \partial z)^2]$ suffers from division-by-zero singularities near the LLJ nose where wind shear vanishes ($\partial u / \partial z \rightarrow 0$). SBLToolkit.jl applies Tikhonov-Morozov filtering and Zero-Offset Hyperbolic Regularization (ZOHR) to guarantee bounded numerical updates.



Richardson Number Profile: Raw vs ZOHR Regularized

Unregularized Ri spikes to infinity at the Jet nose (30m), whereas ZOHR maintains smooth, differentiable stability bounds.



TRACK A PRIMITIVE REGULARIZATION

Applies Tikhonov-Morozov filtering directly to velocity and potential temperature profiles under a Modified Morozov Discrepancy Principle (MDP). Matches smoothing scale to noise floor δ .

ZERO-OFFSET HYPERBOLIC SCHEME (ZOHR)

Replaces singular quotient denominators with hyperbolic smoothing operators $\sqrt{x^2 + \epsilon^2}$. Eliminates division-by-zero without altering physical equilibria.

TANGENTIAL CONE CONDITION

Ensures non-linear stability bound $\tau_2 \geq \tau_1(3 + 2\gamma)$, guaranteeing a unique regularization parameter α for non-linear inverse convergence.

5. Quality Control Gating & GSPT Framework Architecture

Triple-Point Dispersion QC validation and multi-phase prognostic GSPT integration.

To prevent silent numerical corruption from creating false regime transitions, SBLToolkit.jl enforces a strict Quality Control pipeline using Triple-Point Dispersion (δ_{TP}) alongside schema adapters (``src/ultra/adapters/``) to sanitize incoming NetCDF tower streams.

1

Diffusivity Cutoff Height

Vertical height where turbulent eddy diffusivity K_m drops below physical mixing threshold ($10^{-4} \text{ m}^2/\text{s}$).

Marker $z_K : K_m(z) \rightarrow \epsilon$

2

TKE Level-Set Height

Height level where Turbulent Kinetic Energy drops below critical threshold (e.g. $e = 0.05 \text{ m}^2/\text{s}^2$).

Marker $z_e : \text{TKE}(z) = e_{\text{crit}}$

3

TKE Gradient Extremum

Point of maximum TKE vertical collapse gradient ($\partial e / \partial z = \max$), marking active collapse front.

Marker $z_{\text{grad}} : \max(|\partial \text{TKE} / \partial z|)$



GSPT Evolutionary Phases: Diagnostic Mapping to Prognostic Closure

PHASE 1

Data Cleansing & Regularization

Planar fit rotations, u- flatline replacements (0.3 m/s baseline), and Track A primitive regularization.

PHASE 2

Diagnostic Inversion (R_{coord})

Maps profiles onto topological manifold coordinate R_{coord} to distinguish shear mixing from laminar collapse.

PHASE 3

Prognostic Closure Integration

Embeds R_{coord} manifold geometries into 1.5-order & 2nd-order turbulence models for jet shear revival prediction.